

1. Knowledge QA

A reviewer can ask a COF-related question in natural language and inspect how the answer is grounded in graph and literature evidence.

COF-System AI4S
v1.0.0

知识问答

COF 设计分析

单体库

系统设置

知识问答

自然语言问题、图检索、知识片段、可审计分析步骤

自然语言输入

Schiff-base 和 beta-ketoenamine 两种方式对 COF 的光催化析氢表现有什么影响?

发送并检索

分析步骤

6/6 done

问题归类

done

实体解析

done

知识图检索

done

图检索

done

证据组织

done

最终结论

done

最终答案

Schiff-base 与 β -ketoenamine 两种连接方式对 COF 的光催化析氢性能有显著影响。Schiff-base 反应形成的可逆共价键为构建有序框架提供了动态调控窗口,而不可逆的 β -ketoenamine 结构则赋予材料更稳定的化学稳定性。研究表明,通过羧基功能化修饰 β -ketoenamine-linked COF,可有效抑制光降解并延长载流子寿命,使析氢量子产率提升至 37% (未修饰体系仅 7.8%)。此外,调节 β -ketoenamine 与羧基的比例能优化框架有序度,增强可见光吸收能力,但过度增加刚性结构可能导致比表面积下降,反而削弱传质效率。目前证据表明,两种连接方式的协同作用可通过分子工程手段走向调控,但具体构效关系仍需更多实验验证。

图检索证据

关系	用实例
Electrodes	INVOLVE COF
environmental issues	INVOLVE COF
NF-kappa B activation	INVOLVE COF
copper coordination environment c...	INVOLVE COF
cleavage of H-Cl covalent bond	INVOLVE COF
coordination bond cleavage (Cu-N)	INVOLVE COF
structural microenvironment	INVOLVE COF
synthetic_methods	INVOLVE COF
mandible	INVOLVE COF
maxilla	INVOLVE COF
aqueous media	REACTION_CONDITIONS COF
phenol degradation	PHOTOCATALYTIC_ACTIVITY COF
catalyst	REUSABLE COF
for catalysis	REUSABLE COF
multiple reaction cycles	REUSABLE COF
PC-TpPa-1	MODIFIED_TO COF
MCOF	MODIFIED_TO COF
COF-CN	MODIFIED_TO COF

原始子图 JSON (已展开)

```
{
  "source": "nextj",
  "nodes": [
    {
      "id": "schiff-base Electrodes",
      "label": "Electrodes",
      "label": [
        "Result"
      ],
      "properties": {
        "value": "Electrodes"
      }
    },
    {}
  ],
  "edges": [
    {}
  ]
}
```

Cypher 查询 (已展开)

```
MATCH (a:Node)-[r]-(b:Node)
WHERE toLower(a.name) CONTAINS toLower("schiff") OR toLower(a.name) CONTAINS toLower("base") OR
toLower(a.name) CONTAINS toLower("schiff") OR toLower(a.name) CONTAINS toLower("base")
OR toLower(a.name) CONTAINS toLower("beta-ketoenamine") OR toLower(a.name) CONTAINS toLower("beta-
ketoenamine")
OR toLower(a.name) CONTAINS toLower("cof") OR toLower(a.name) CONTAINS toLower("cof")
OR toLower(a.name) CONTAINS toLower("photocatalytic") OR toLower(a.name) CONTAINS
toLower("photocatalytic")
OR toLower(a.name) CONTAINS toLower("hydrogen") OR toLower(a.name) CONTAINS toLower("hydrogen")
RETURN a.name AS source, type(r) AS relation, b.name AS target
LIMIT 100
```

知识库片段

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KB-1

[KB-1] Activation of Carbonyl Oxygen Sites in β -Ketoenamine-Linked Covalent Organic Frameworks via Cyano Conjugation for Efficient Photocatalytic Hydrogen Evolution

1. Introduction · DOI: 10.1002/ami.202101017

Among various COFs, β -ketoenamine-linked COF, a class of $S-C(-N-)$ covalent frameworks, was first reported by Banerjee et al.[16] The preparation processes proceed with reversible Schiff-base reaction to construct an enol form with crystalline arrays and then irreversible enol-keto tautomerism to get the final keto-form networks, which exhibit the extraordinary chemical stability in boiling water, as well as in acidic H_2O and basic solutions $NaOH$ and basic solutions $NaOH$. Benefiting from these features, the ketoenamine-based materials have quickly evoked soaring interest in photocatalysis research field. The previous works were mainly committed to the morphology control, chemical stability, and charge transport of the COFs[17] Though the photocatalytic performance of β -ketoenamine-linked COFs has been improved to some e...

INPUT

Submit a natural-language COF question.

ANSWER

Read the synthesized response after retrieval.

TRACEABILITY

Inspect graph triples, Cypher, raw JSON, and paper fragments.

USE CASE

Use this page when the reviewer needs to see how COF knowledge questions are answered and traced.

COF-System AI4S reviewer demo PDF

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The design page evaluates whether a monomer combination is chemically plausible for COF construction and surfaces the evidence chain behind the recommendation.

Video example: Tp + pPD/Pa, imine linkage, Schiff-base conditions, 3D preview, and an eight-step evidence-supported analysis.

3. Monomer Library and 3D Preview

The monomer library supplies curated input records for design analysis and lets users inspect structures before using them in a workflow.

The screenshot displays the COF-System AI4S v1.0.0 interface. On the left, a sidebar contains navigation links: 知识库, COF 设计分析, 单体库 (selected), and 系统设置. The main area is titled '单体库' (Monomer Library) and includes a search bar and filters for Role, Functional group, Source, Review status, and 3D status. A list of monomers is shown, with 'TA, PDA, BDA' selected. The right panel, titled '单体详情 (不可编辑)' (Monomer Details - Not Editable), shows fields for Name, IUPAC, SMILES, Canonical SMILES, Functional Groups, and Composable COF. Below this, a '单体 3D 预览' (Monomer 3D Preview) section shows a 3D ball-and-stick model of the selected monomer. At the bottom right, a '可合成 COF / 二聚体预览' (Synthesizable COF / Dimer Preview) table lists various COFs and their synthesis conditions.

COF	共聚单体	连接键	反应 / 性能	状态
CTF-HUST-1	benzene-1,4-dicarboxylic acid	auto	Schiff base formation f...	可
g-C18N3-COF	TMT	auto	Knoevenagel condensa...	可
ANW-1	TAPB	auto	Schiff base formation f...	可
OB-PCOF-1	2,4,6-trimethylbenzene...	auto	Knoevenagel condensa...	可
COF-DOB	TMBP	auto	Knoevenagel condensa...	可
COF-imine	TFPB	auto	Schiff base formation f...	可

SEARCH

Search by keyword and filter by role, group, source, or 3D status.

INSPECT

Review identifiers, metadata, groups, and related records.

PREVIEW

Generate 3D views, switch styles, and inspect dimers.

ROLE IN THE WORKFLOW

This interface provides the curated building-block layer for downstream COF design analysis.